

Assignment-Regression Algorithm

1. Identify your problem statement

A client's requirement is he wants to predict the insurance charges based on the input data (age, sex, bmi, children, smoker).

Problem Identification:

Stage 1: Machine Learning

Stage 2: Supervised

Stage 3: Regression

2. Tell basic info about the dataset (Total number of rows, columns)

Total no. of rows = 1338

Total no. of columns = 6

3. Mention the pre-processing method

We have the data as string in sex and smoker columns. Since both are ordinal data, we should convert it into numbers using getdummies function.

4. Final model with best r2_score

Random Forest Algorithm develops the best model with R2 score as 0.87,

- When Criterion is squared_error, Max Features is log2 and N_Estimators is 100
- When Criterion is absolute_error, Max Features is sqrt and N_Estimators is 100
- When Criterion is absolute_error, Max Features is log2 and N_Estimators is 100
- When Criterion is friedman_mse, Max Features is log2 and N_Estimators is 100

Please find the below details of R2 score value with different algorithms.

Finally, we take Random Forest Algorithm to create the best model for the given dataset.

Hands-on for finding the best R2 value

1. Multiple Linear Regression:

R2 value is 0.79

2. Support Vector Machine Regression:

		R-value			
S.No	Regularization parameter	Linear	Poly	RBF	Sigmoid
1	C=10	0.46	0.04	-0.03	0.03
2	C=100	0.62	0.62	0.32	0.52
3	C=500	0.76	0.82	0.66	0.44
4	C=1000	0.76	0.85	0.81	0.28
5	C=2000	0.74	0.86	0.85	-0.59
6	C=3000	0.74	0.85	0.86	-2.12

When C=2000 and kernel is poly, R2 value is 0.86

When C=3000 and kernel is rbf, R2 value is 0.86

3. Decision Tree:

S.No	Criterion	Max Features	Splitter	R Value
1	squared_error	None	best	0.68
2	squared_error	None	random	0.74
3	squared_error	sqrt	best	0.53
4	squared_error	sqrt	random	0.69
5	squared_error	Log2	best	0.68
6	squared_error	Log2	random	0.70
7	absolute_error	None	best	0.68
8	absolute_error	None	random	0.71
9	absolute_error	sqrt	best	0.72
10	absolute_error	sqrt	random	0.66
11	absolute_error	Log2	best	0.63

12	absolute_error	Log2	random	0.68
13	Friedman_mse	None	best	0.70
14	Friedman_mse	None	random	0.74
15	Friedman_mse	sqrt	best	0.75
16	Friedman_mse	sqrt	random	0.75
17	Friedman_mse	Log2	best	0.66
18	Friedman_mse	Log2	random	0.71

When Criterion is Friedman_mse, Max Features is best/random and Splitter is sqrt, R2 value is 0.75

4. Random Forest Regression

S.No	Criterion	Max Features	N_Estimators	R Value
1	squared_error	None	10	0.84
2	squared_error	None	100	0.85
3	squared_error	sqrt	10	0.84
4	squared_error	sqrt	100	0.86
5	squared_error	Log2	10	0.85
6	squared_error	Log2	100	0.87
7	absolute_error	None	10	0.86
8	absolute_error	None	100	0.85
9	absolute_error	sqrt	10	0.86
10	absolute_error	sqrt	100	0.87
11	absolute_error	Log2	10	0.85
12	absolute_error	Log2	100	0.87
13	friedman_mse	None	10	0.84
14	friedman_mse	None	100	0.85
15	friedman_mse	sqrt	10	0.85
16	friedman_mse	sqrt	100	0.86

17	friedman_mse	Log2	10	0.85
18	friedman_mse	Log2	100	0.87

When Criterion is squared_error, Max Features is log2 and N_Estimators is 100, R2 value is 0.87

When Criterion is absolute_error, Max Features is sqrt and N_Estimators is 100, R2 value is 0.87

When Criterion is absolute_error, Max Features is log2 and N_Estimators is 100, R2 value is 0.87

When Criterion is friedman_mse, Max Features is log2 and N_Estimators is 100, R2 value is 0.87